

Example Document: LaTeX Docker Environment

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Abstract

This is an example document to verify that the LaTeX Docker environment is working correctly. It demonstrates several common LaTeX features including math, figures, tables, code listings, and bibliography management with `biblatex` and `biber`.

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1 Introduction

Welcome to your LaTeX Docker environment! This document serves as both a verification test and a reference for common LaTeX constructs.

1.1 What This Environment Provides

This setup includes:

- **Full TeX Live distribution** — every package available
- **Multiple engines** — pdfLaTeX, XeLaTeX, LuaLaTeX
- **Automated builds** — via `latexmk` and Docker Compose
- **VS Code integration** — with LaTeX Workshop extension
- **Bibliography support** — using `biblatex` + `biber`

1.2 Getting Started

To compile this document, run:

```
1 ./compile.sh example
```

Or create a new project:

```
1 ./compile.sh new my-project
```

As shown in the literature [1, 2], L^AT_EX remains the gold standard for scientific typesetting.

2 Mathematics

L^AT_EX excels at typesetting mathematical content.

2.1 Inline and Display Math

Euler's identity $e^{i\pi} + 1 = 0$ is often considered the most beautiful equation in mathematics.

The quadratic formula solves $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

We can reference eq. (1) easily with `cleveref`.

2.2 Aligned Equations

Maxwell's equations in differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad (2)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (3)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (5)$$

2.3 Theorem Environments

Theorem 2.1 (Pythagorean Theorem). *For a right triangle with legs a , b and hypotenuse c :*

$$a^2 + b^2 = c^2$$

Proof. The proof is left as an exercise for the reader. □

2.4 Matrices

A rotation matrix in 2D:

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (6)$$

3 Figures and Tables

3.1 Tables

Table 1 compares the L^AT_EX engines available in this environment.

Table 1: Comparison of L^AT_EX engines

Engine	Font Support	Speed	Best For
pdfLaTeX	Type1, OTF (limited)	Fast	Most documents
XeLaTeX	OpenType, TrueType	Medium	Custom fonts, Unicode
LuaLaTeX	OpenType, TrueType	Slower	Lua scripting, fonts

3.2 Figures

To include a figure, place your image in the `figures/` directory:

```
\begin{figure}[H]
  \centering
  \includegraphics[width=0.8\textwidth]{figures/my-image.pdf}
  \caption{Description of figure}
  \label{fig:example}
\end{figure}
```

4 Conclusion

If this document compiled successfully, your $\text{\LaTeX}$ Docker environment is fully operational. You now have access to:

1. The complete TeX Live distribution with all packages
2. Three compilation engines: pdfLaTeX, XeLaTeX, and LuaLaTeX
3. Automated compilation via `latexmk`
4. Bibliography management with `biblatex` and `biber`
5. VS Code integration with the LaTeX Workshop extension

Happy typesetting!

References

- [1] Donald E. Knuth. *The T_EXbook*. Reading, Massachusetts: Addison-Wesley, 1984.
- [2] Leslie Lamport. *ΛT_EX: A Document Preparation System*. 2nd. Reading, Massachusetts: Addison-Wesley, 1994.